

SIGCHI Outstanding Dissertation Award: Profiling Artificial Intelligence as a Material for User Experience Design

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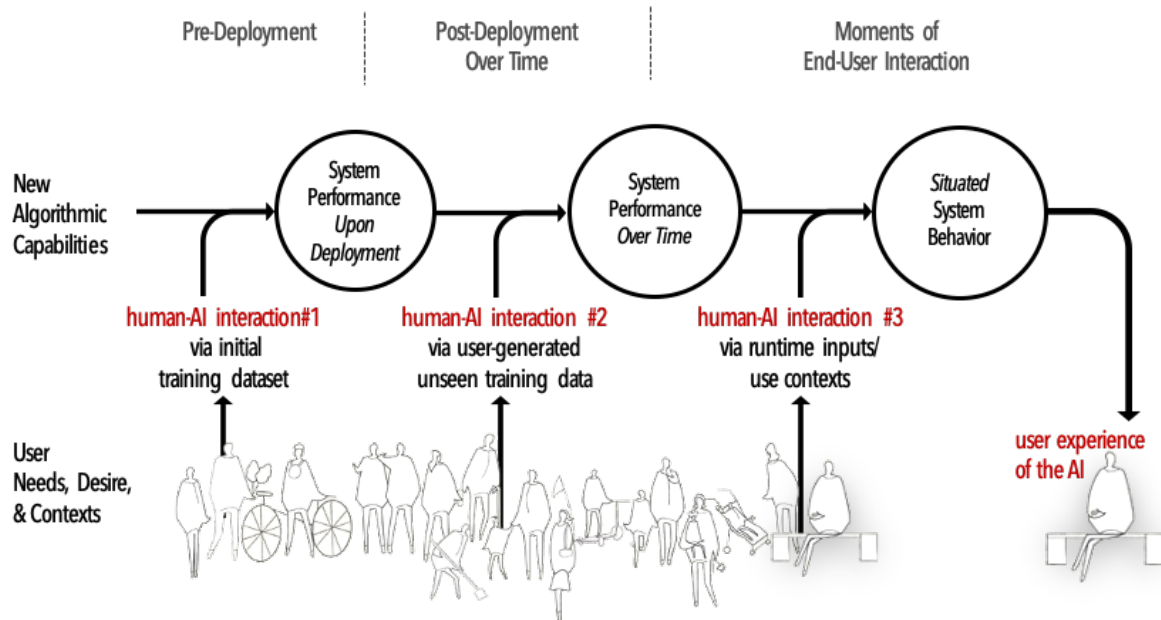


Figure 1: The lifetime of a data-driven machine learning system and its three touch points with people [8]. I distinguish these three types of “human-AI interaction”, the designs of which together shape the user experience of the system.

ABSTRACT

HCI has become especially interested in the promises and challenges of user experience of AI, such as user acceptance, human-agent teamwork, and accessibility. Less discussed, however, is that the field of HCI routinely grapples with such challenges; across the various technologies commonly referred to as AI (e.g., predictive modeling, computer vision, NLP), what shared characteristics made human-AI interaction appeared uniquely difficult to design in the first place? Synthesizing my hands-on design and research over the past six years, in my dissertation, I worked to articulate whether, why, and how human-AI interaction appears uniquely challenging to design with established HCI methods. In this extended abstract, I first describe a human-AI interaction design framework as an

answer to this question. I then discuss one critical implication of this framework: Framing data-driven AI systems as living socio-technical systems that co-evolve with their users. I analogize this reframing to the shift from desktop computing to ubiquitous computing and outline the ethnographic, design, and technological research opportunities it reveals.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction design process and methods*.

KEYWORDS

User experience, artificial intelligence, sketching, prototyping.

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1 PROFILING USER EXPERIENCE OF AI

Advances in Artificial Intelligence (AI) offer both opportunities and challenges for user experience (UX) design. From mundane spam filters, to conversational agents at home, to autonomous driving, AI systems promise to improve human-computer interaction (HCI) and enable otherwise impossible forms of interaction. Meanwhile, these systems can make inference errors that are difficult to anticipate, and such errors can damage UX and even lead to undesired societal impact. For example, AI systems can make unpredictable errors, and these errors damage UX and even lead to undesired societal impact. However, HCI routinely grapples with complex technologies that make unexpected errors, and mitigates their unintended consequences. – *What makes AI different?* What makes addressing AI's perils appear particularly challenging [1, 5], even with the many HCI methods already under our belt?

This is a difficult question. The research landscape of human-AI interaction remains relatively disparate between its vast and soft goals (e.g. human-AI complementarity) and valuable yet system-or-domain-specific solutions (e.g. improving physician-AI complementarity on a narrow clinical decision, with a particular kind of AI system).

My dissertation is an initial attempt to connect these dots [4]. Over the course of my PhD, I designed a variety of AI applications based on empirical studies of their actual stakeholders and real-world impacts. These applications span the domains of healthcare [6, 10], accessibility [2, 11], productivity [3, 7], social networks, and more. Each design addressed a critical challenge in translating AI from a technological advance to a socio-technical system valuable to human ends. In addition, I investigated how the researchers and industry practitioners leveraged – or failed to leverage – AI in their respective domains of interest. This includes empirical studies with nearly 150 industry practitioners, offering rare and rich industry best practices in crafting human-AI interaction, as well as the persisting challenges [5, 9].

The result of this work is a new framework that delineates when, why, and how the challenges of human-AI interaction are different from general HCI challenges (Figure 1). It highlights the three touch points with people in the lifetime of a data-driven AI system, first via training datasets, then via user-generated unseen training data post-deployment, and finally via various user inputs that allow the system to provide personalized and context-aware outputs. How we design the “human-AI interactions” across these three touchpoints together shape the user experience of the system. Crafting the UX of an AI system is challenging because, across these touch points, the interactions involve 1) three different groups of humans and the human complexities, 2) an AI system constantly evolving and its complexities, and 3) the semi-autonomous mutual adaptation between the two over time.

Consider the challenge of *accounting for algorithmic biases in human-AI collaborative decision making*, a challenge that AI systems face in many critical domains such as healthcare and criminal justice. Building a “fair” AI application is widely considered as difficult, due to the complexity both in defining fairness goals and in algorithmically achieving the defined goals. This framework further illuminates that building “a fair AI system pre-deployment” is only part of the whole “AI fairness” problem space (Figure 2).

Take intelligent clinical-decision support systems as an example. How might clinicians and the AI respectively adapt to each other's behaviors over time, and how might the mutual adaptation affect fairness? Previous field work [6] has revealed that, for some cardiac surgeries that have officially defined AI models used to rate surgeons and care teams, clinicians' decision meetings had become centered around risk models. One surgeon said: *“It's not that we don't help that patient, but if we take this shot and do poorly, then we cannot take on the next 10 patients like him. [...] Insurance companies and Medicare and all that... they will mark you. They may not pay. It all plays into the complex factor for deciding who, especially sicker patients, we would take a shot.”* AI systems, by way of the algorithmic simulation and prediction of “good” clinical decisions, can shift clinician behaviors and alter patient futures. These effects could take months if not years to detonate, yet raise critical questions worth consideration from the very beginning of the AI design and development process.

2 AI AS A MATERIAL FOR USER EXPERIENCE DESIGN

Modern AI systems are living systems that can co-evolve with their human contexts. It is in this light that I analogize designing human-AI interaction to designing ubiquitous computing interactions: Both break away from the convention of designing a more user-friendly AI system/desktop computer, and instead systematically reshape AI/computation to better fit in people's day-to-day lives. Both mark an HCI paradigm shift that calls for:

- (1) New *ethnographic and design work* that re-imagines when, where, and how people may perceive AI/computation as valuable;
- (2) A wave of *technological and interaction design innovations* that together enable AI/computing devices to deliver value when, where, and how it is needed;
- (3) A renewed understanding of *what “AI”/“computer” is and means* for people and for societies at large;
- (4) New *practitioner-facing processes and tools* that integrate and transfer the above innovations and insights to practicing communities, thereby materializing AI/computation's impact in everyday technology products and services.

This dissertation detail two of the hands-on projects as case studies, exemplifying how researchers can effectively broach into AI's UX design opportunities and challenges, by distinguishing how they relate to and differ from UX design in general. The case studies, combined with the proposed framework, provide a first step towards connecting islands of prior research and building a smoother pathway between AI's technological advances and valuable AI-enabled user experiences.

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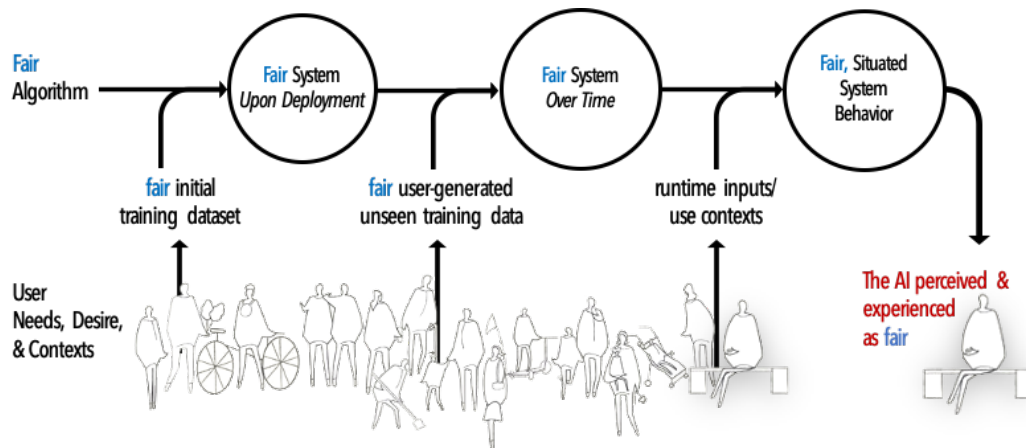


Figure 2: Mapping the challenges of “AI fairness” using the framework.

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3 SPEAKER’S BIO

Qian Yang is an assistant professor at Cornell University Ann S. Bowers College of Computing and Information Science. She received her Ph.D. from Carnegie Mellon University HCI Institute. Her work focuses on integrating Artificial Intelligence applications (e.g., predictive modeling, computer vision) into human-centered technology design practices. Taking a Research through Design approach, Yang investigates how to better design human-AI interaction by designing them: from decision support systems for life-and-death healthcare decisions to intelligent mobile services, from Natural Language Generation systems to autonomous cars. Yang’s AI system designs have won multiple awards both from academic conferences and industry.

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